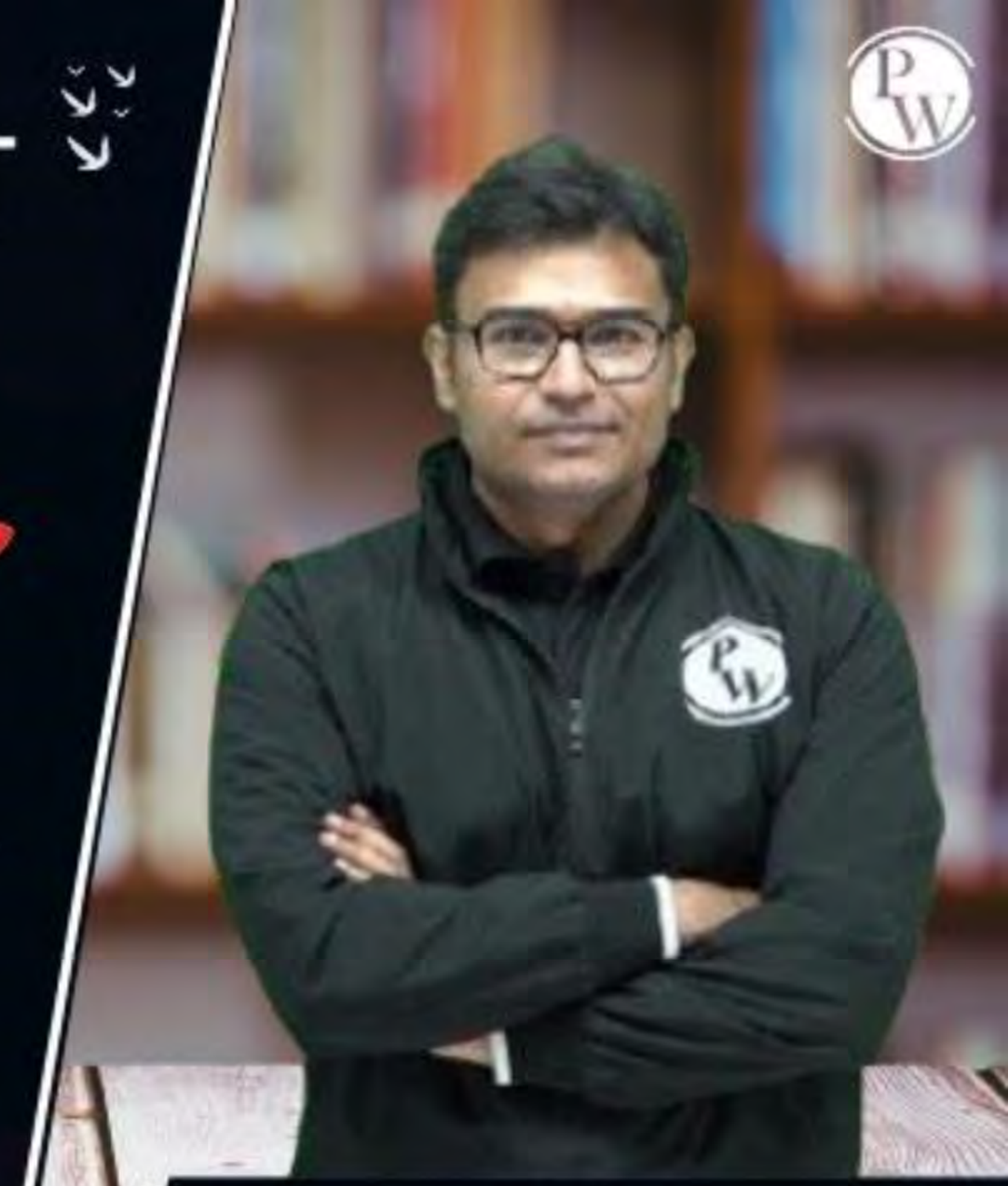
Computer Science & IT

Theory of Computation



CFL

Lecture No. 4



By- Ravindra Sir

Recap of Previous Lecture



Topic

Topic

Topic

DCFL, CFL

1000 qus

Compulsory

12:30

9-11:30 → CFL

Optional

11:30-12:30

Regular PYQ + practice questions

Topics to be Covered



- Topic
- Topic
- Topic

CFL

eliminate ∈ productions

eliminate unit production

eliminate useless productions

CNF, GNF → CYK → CFGs → parsing

def x

ab → 3 states

ab → 4 states

$a^n b^n$ $S \to aSb/ab$

Peter Linz

ullman

11:30-12:30

$a^{n^2} / n \geq 1$ Regx CFLX CSL✓

$a^{2^n} / n \geq 1$ Regx CFLX CSL✓

$a^{n!} / n \geq 1$ Regx CFLX CSL✓

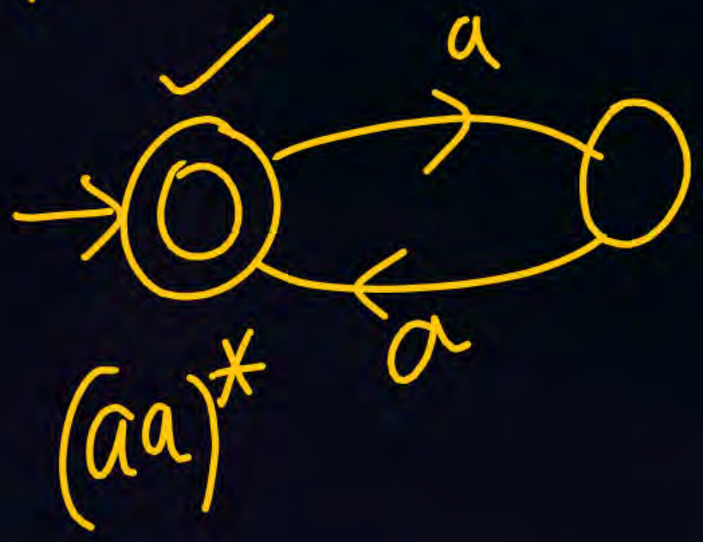
a^m / m is prime Regx CFLX CSL✓

x finite

a^k / k is even → Regular

$a^{(k+1)}$ / k is even → a^l / l is odd → Reg

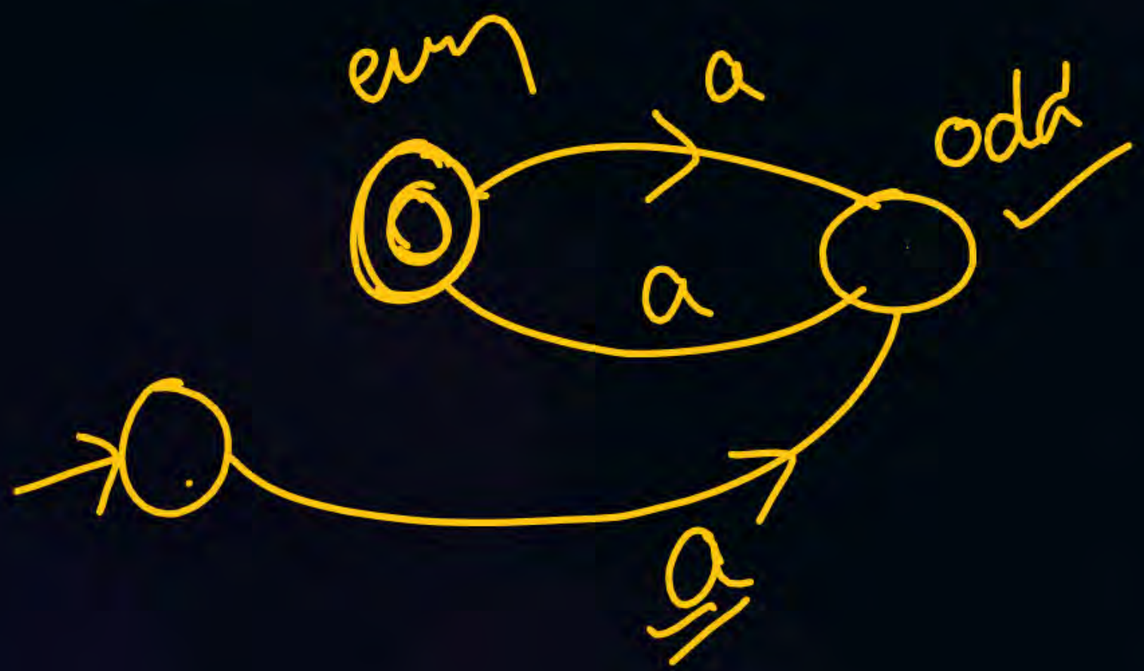
a^k / k is even



a(aa)*

(aa)*a

$(aa)(aa)^*$



$a^{k/2}$ ✓ k is even ✓

$a^{0/2}, a^{2/2}, a^{4/2}, a^{6/2}, a^{8/2} \ldots\ldots\ldots$

$a^0, a^1, a^2, a^3 \ldots\ldots$

Reg lang → finite
→ infinite



$a^i b^j c^k \; / \; i > k$

Push a — no action — pop a

final state $(\in, \in / a)$

$a^i b^j c^k$ / $i > k$ ✓DCFL ✓TFL ✓CSL

a → no action
b → push
c → pop b

final → $(\epsilon, b/b)$

$a^i b^j c^k / i > j > k$ CSL ✓

Regx DCFLx CFLx

$a^i b^j c^k / j = i+k$

$a^i b^{i+k} c^k$

$a^i b^i b^k c^k$

$\underline{a}\ \underline{b}\ \underline{b}\ \underline{c}$

$P_1\ P_0\ P_1\ P_{0A}$

$a^i b^j c^k d^l / i=k \& j=l$

$a=c$ ✓

$b=d$ ✓

✓ CSL

✓ CFL

Regx

DCFLX

NPDA

$a=c$

$b=d$

$a^i b^j c^k d^l / i=k$ and $j=l$

Reg ✗

DCFL ✗

CFL ✗

CSL ✓

$a^m b^l c^k d^n / m, l, n, k \geq 1$

$a^+ b^+ c^+ d^+ \rightarrow RE$

↓

- Regular
- ✓ DCFL
- ✓ CFL
- ✓ CSL

$a^n b^{4m} / n, m \geq 1$

$aa^* bbbb(bbbb)^*$

$a^+ (bbbb)^+ \rightarrow RE \rightarrow Reg$

$a^n b^{4n} / n \geq 1 \rightarrow dCFL$ ✓

$a^n b^{kn} / k$ is a positive constant $\rightarrow$ RegX $\rightarrow$ DCFL

$a^{kn} b^n / k$ is a fixed constant

↓ DCFL

$\{a^3, a^8, a^{13}, a^{18} \ldots\ldots\ldots\} \rightarrow$ Reg

$aaa(aaaaa)^*$

$\{a^{2n+1} / n \geq 1\} \rightarrow$

aaa, aaaaa ----

$aaa(\underline{aa})^* \rightarrow$ RE $\rightarrow$ Reg

$a^n / n \geq 1$ Reg ✗ CFL ✗ CSL ✓

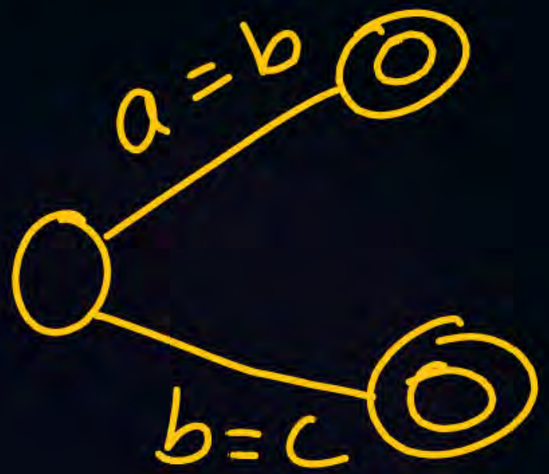
$ww / w \in (a,b)^*, |w| = 100 \rightarrow$ finite $\rightarrow$ Reg

$w / w \in \{a,b\}^*, n_a(w) = n_b(w) \rightarrow$ DCFL

✓ $ww / w \in (a,b)^* \rightarrow$ CSL

$w / w \in \{a,b,c\}^*, n_a(w) = n_b(w) = n_c(w) \rightarrow$ PDA ✗ CSL ✓

$w / w \in \{a,b,c\}^*, n_a(w) = n_b(w) \cap n_b(w) = n_c(w)$

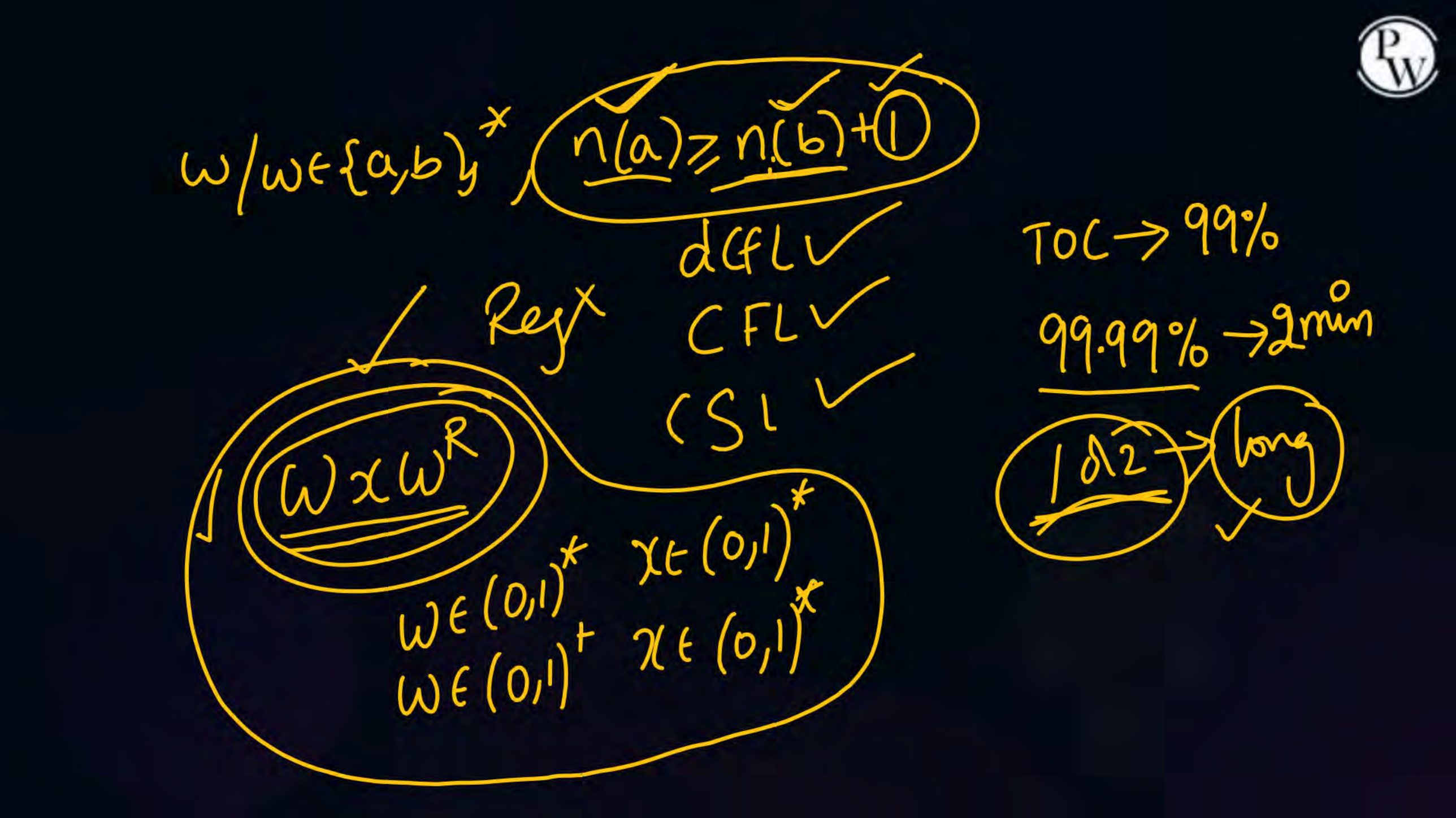


CSL ✓

NDPA → CFL

XDCFL

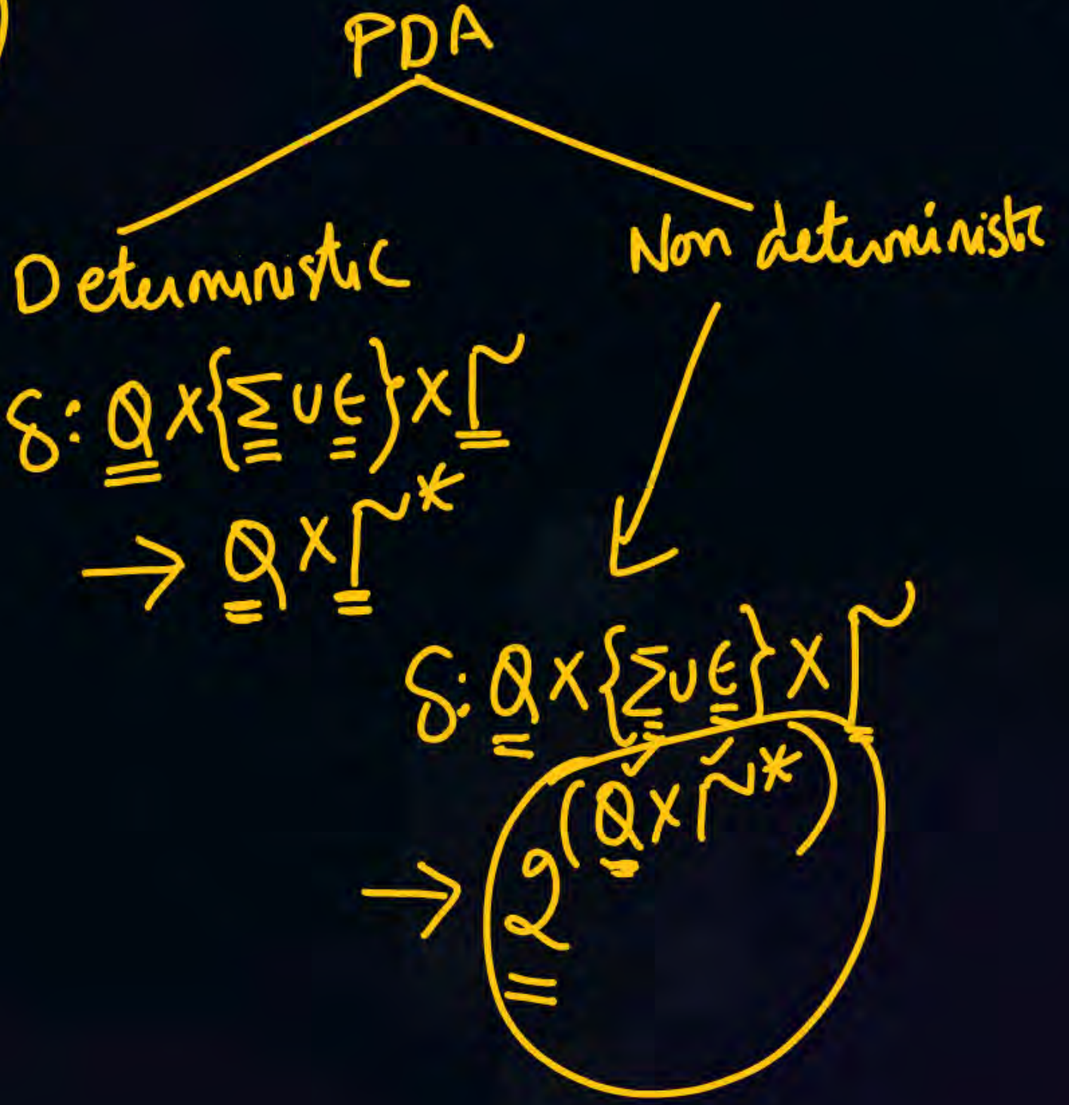
XReg


w/w∈{a,b}*
n(a) ≥ n(b)+1
dCFL ✓
Reg ✗
CFL ✓
CSL ✓
wxw^R
w∈(0,1)*
x∈(0,1)*
w∈(0,1)+
x∈(0,1)*
TOC → 99%
99.99% → 2min
1 & 2
long

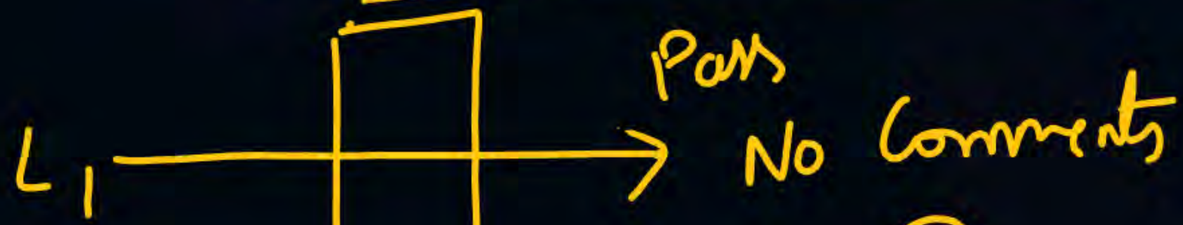
PDA definition:-

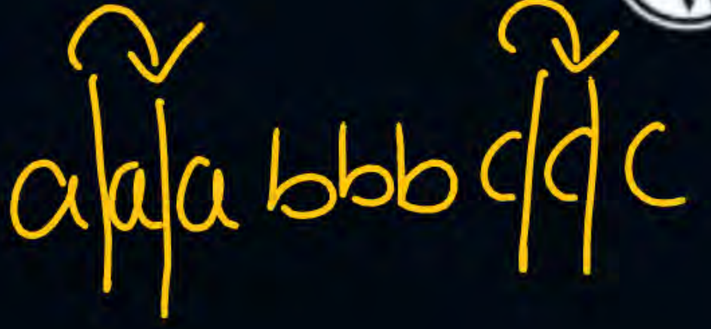
PDA → $(Q, \Sigma, \delta, q_0, Z_0, F, \Gamma)$

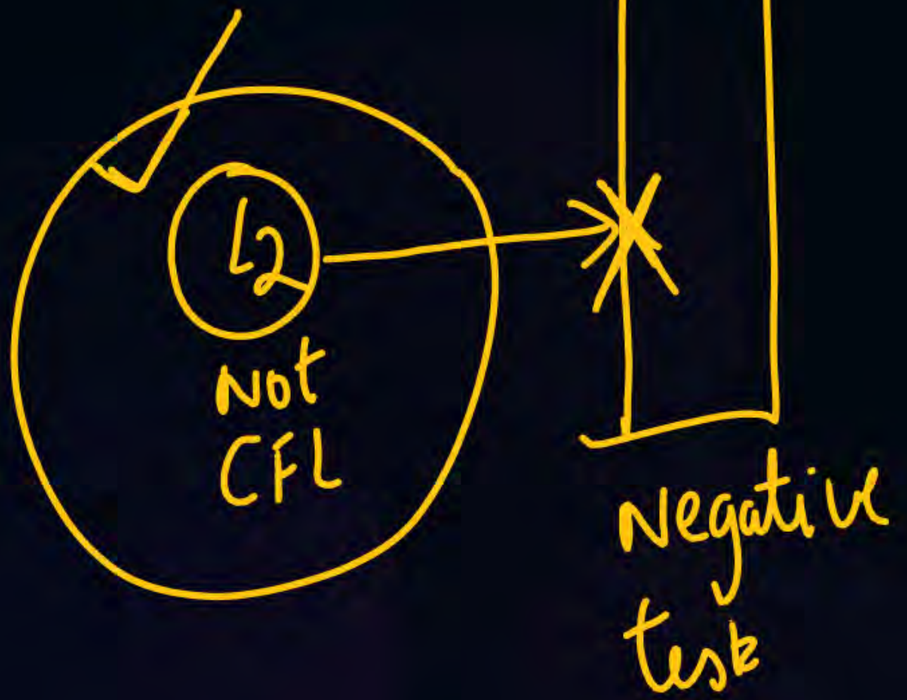
- Q → finite set of states
- Σ → i/p symbols
- δ → Transition function
- q_0 → initial state
- F → Set of final states
- Γ → Stack alphabet
- Z_0 → Bottom of the stack

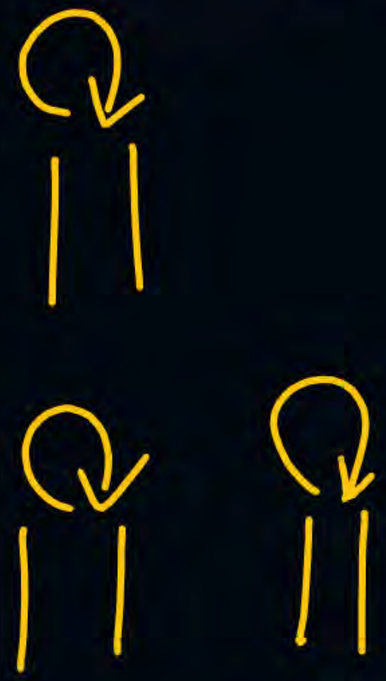


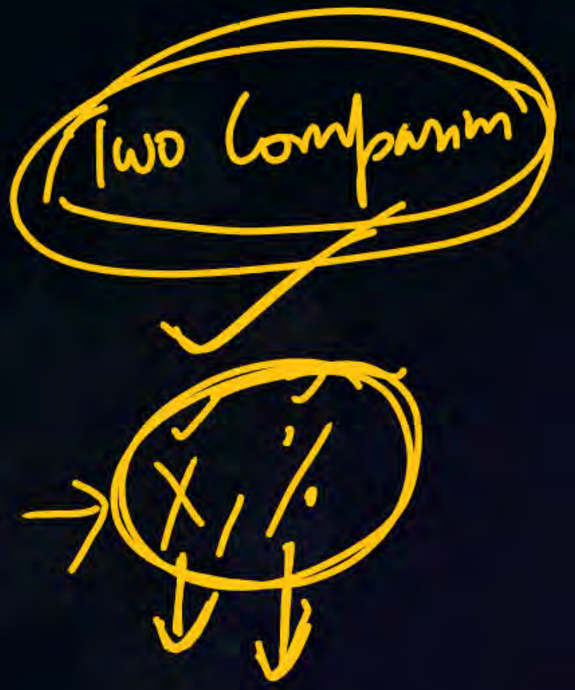
✓ Pumping lemma for CFLS











Pumping lemma for CFL's :-

Let L be a CFL and $w \in L$ such that $|w| \geq n$ for some positive integer 'n', then 'w' can be decomposed as $w = abcde$ such that

(i) $bd \neq \epsilon$

(ii) $|bcd| \leq n$

(iii) for all $i > 0$, strings $ab^i cd^i e$ is also in L



$a^n b^n / n \geq 1$

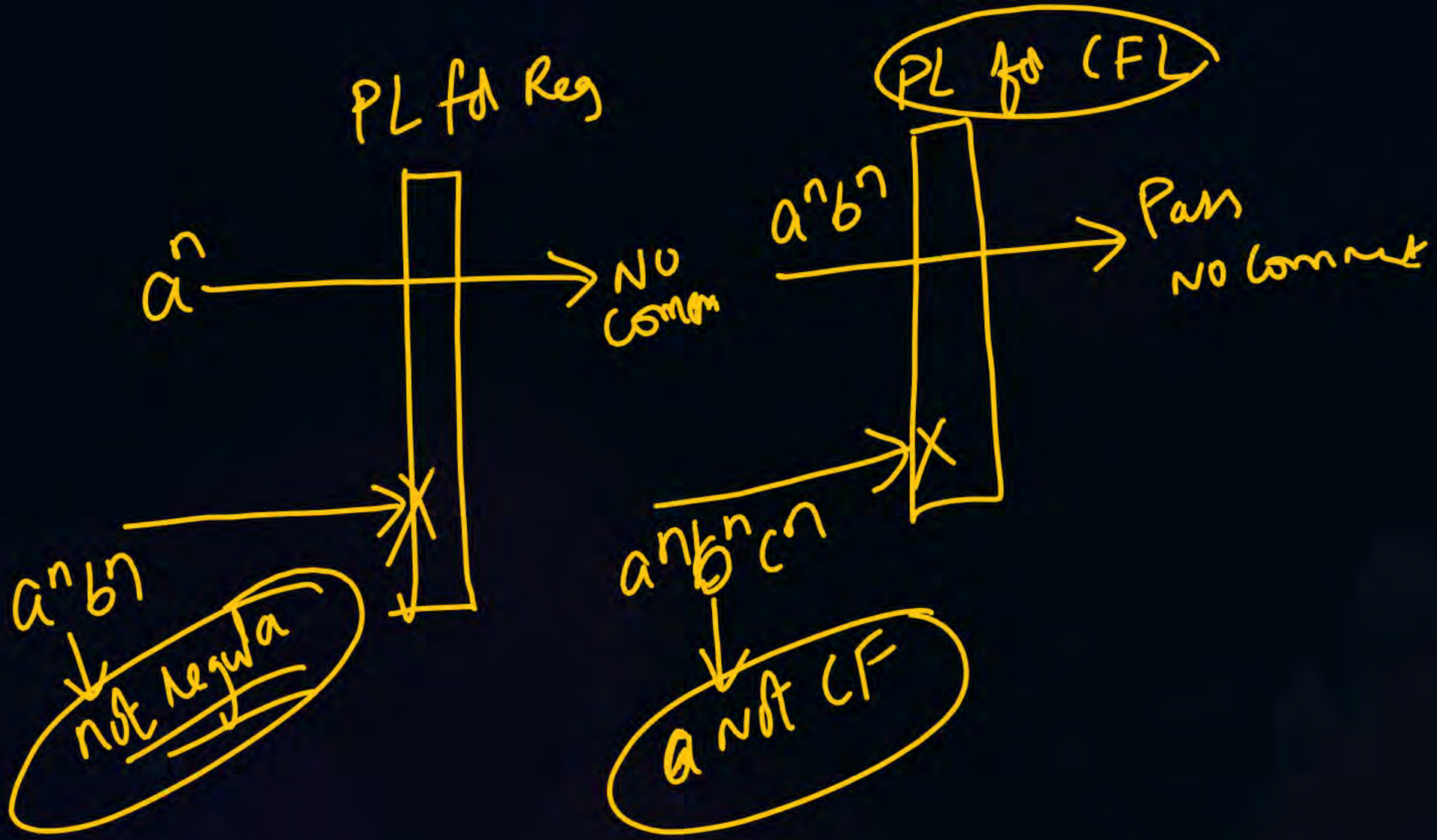
a | a | a | a b | b | b | b → pass PL → No comments

$L_1 \rightarrow$ PL for RL $a^n b^n c^n / n \geq 1$

a | a | a | aa b bbbb | c | c | ccc

L_1 NO Comment

PL → fail → Not CF


PL for Reg
PL for CFL
a^n
NO
$a^n b^n$
$a^n b^n$
Pass
NO
$a^n b^n$
not
$a^n b^n c^n$
a not CF

Context free grammars

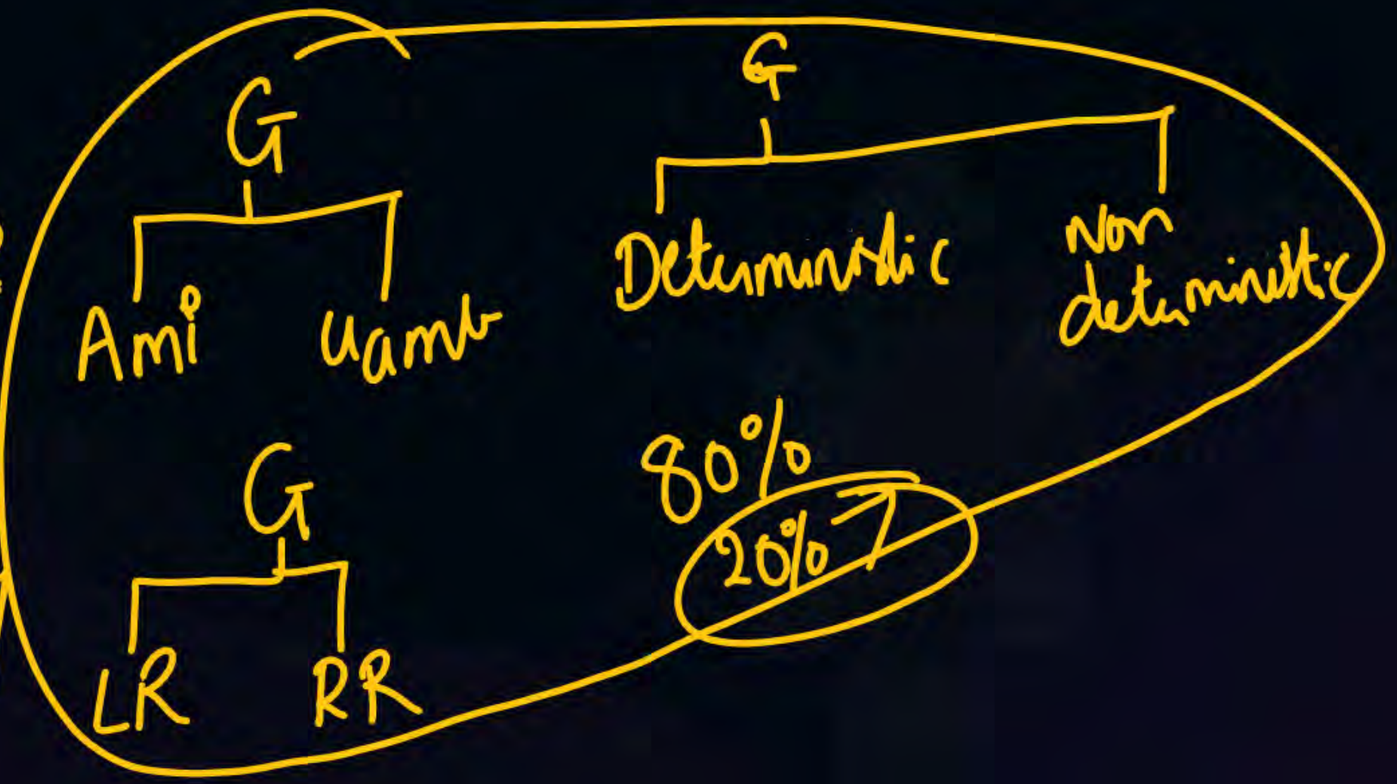
Type 2

$A \to \alpha$ $\begin{pmatrix} A \in \text{Vertices} \\ \alpha \in (V \cup T)^* \end{pmatrix}$

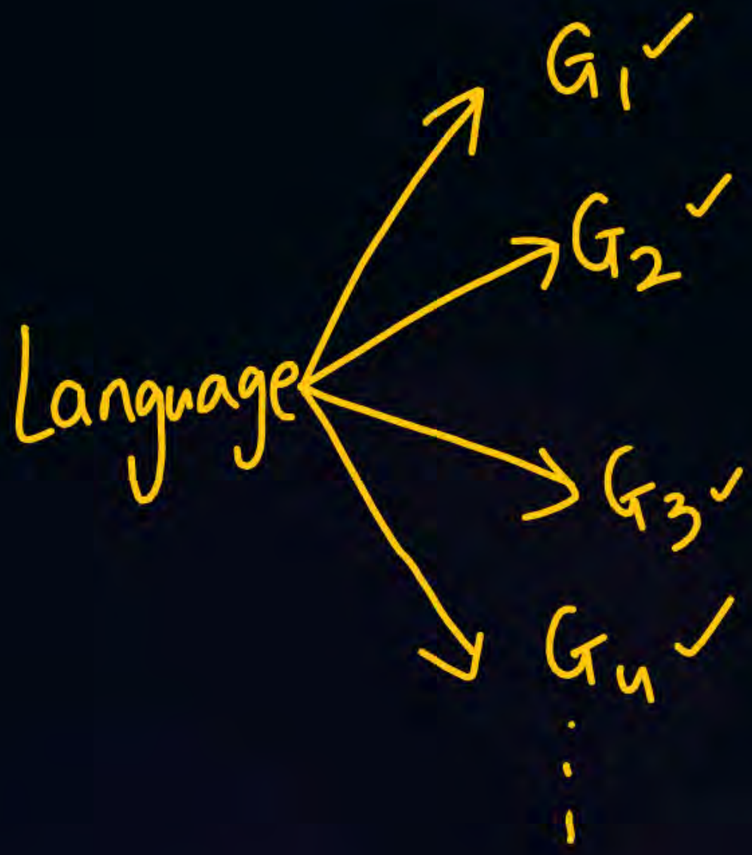
$A \to aA/b$

$V = \{A\}$

$T = \{a, b\}$



A language can have many grammars.
But some grammars are more useful for derivation purposes



$(A \to \epsilon)$ is called ϵ production.

If a language doesn't contain 'ϵ', we can eliminate all ϵ productions.

If a language contains ϵ, then we cannot eliminate all ϵ productions. There will be one production of form $S \to \epsilon$



Jay Bansal
→ 2018 → AIR 2 → MTech-IITB
→ Google-Micro

Elimination of ∈ productions :-

$S \to aSb / aAb$ ✓

$A \to \epsilon$ ✗

If S is not nullable, the language doesn't contain 'ϵ'

Nullable variable = {A}

$S \to aSb / \cancel{aAb} / ab$

$A \to X$

$S \to aSb / ab$

$a^n b^n / n \geq 1$

If a language contains 'ϵ', we can eliminate all ϵ-productions except 'S → ϵ'



$S \to AB/\epsilon$

$A \to aAA/\epsilon$

$B \to bBB/\epsilon$

Nullable variables = $\{A, B, S\}$

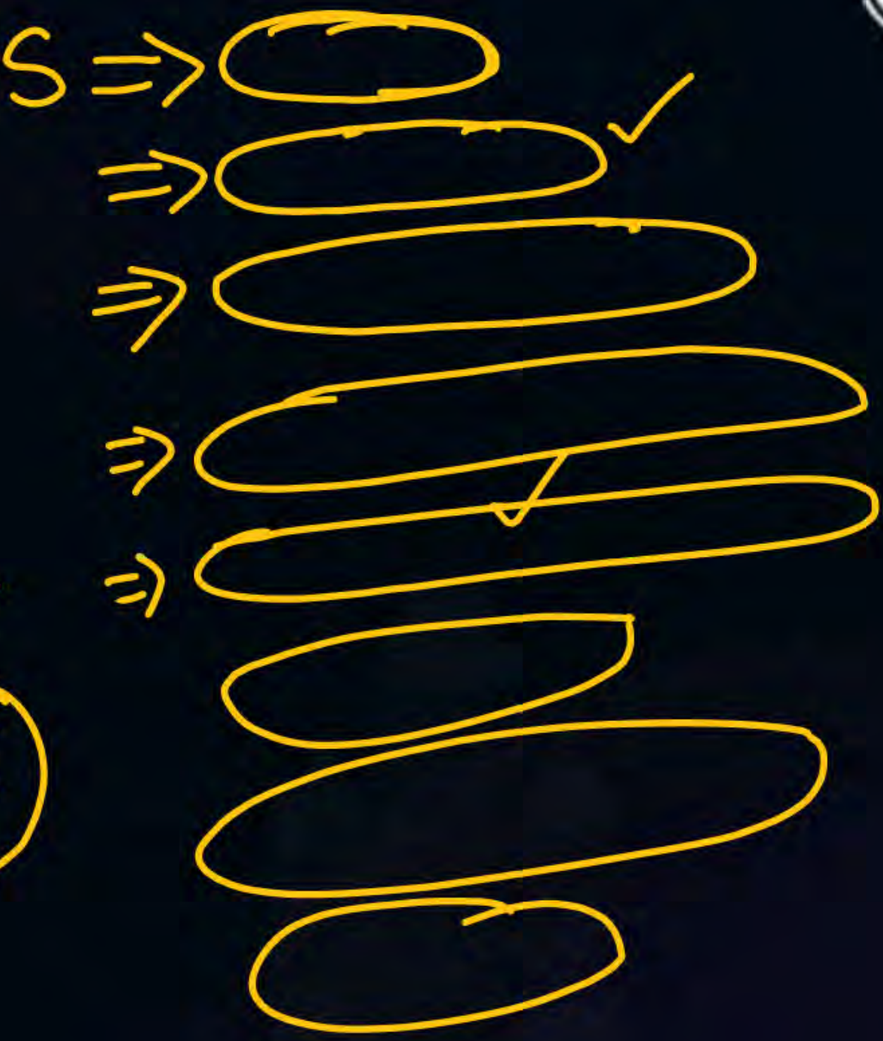
RHS

$S \to AB/\epsilon/B/A$

$A \to aAA/aA/a$

$B \to bBB/bB/b$

$S \to \epsilon$



$S \rightarrow A(ba)C$

$A \rightarrow BC$

$B \rightarrow b / \cancel{\varepsilon}$

$C \rightarrow D / \cancel{\varepsilon}$

$D \rightarrow d$

$S \rightarrow AbaC / baC / Aba / ba$

$A \rightarrow BC / (C) / (B)$

$B \rightarrow b$

$C \rightarrow D$

$D \rightarrow d$

Nullable variables $= \{B, C, A\}$

$A \rightarrow B$

$B \rightarrow C$

$C \rightarrow a$

$A \rightarrow a$

elimination of unit productions :-

$S \rightarrow Aa / B / bb / a / bc$

$B \rightarrow A / bb / a / bc$

$A \rightarrow a / bc / B / bb$

$A \rightarrow B \rightarrow bb / a / bc$

$S \rightarrow B \rightarrow bb$

$S \rightarrow B \rightarrow A \rightarrow a / bc$

$B \rightarrow A \rightarrow a / bc$

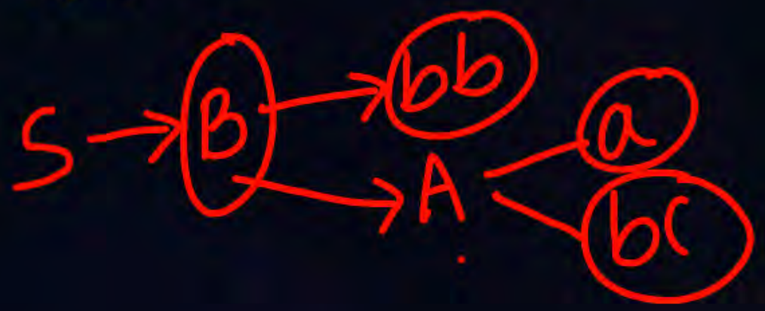
$B \rightarrow bb$

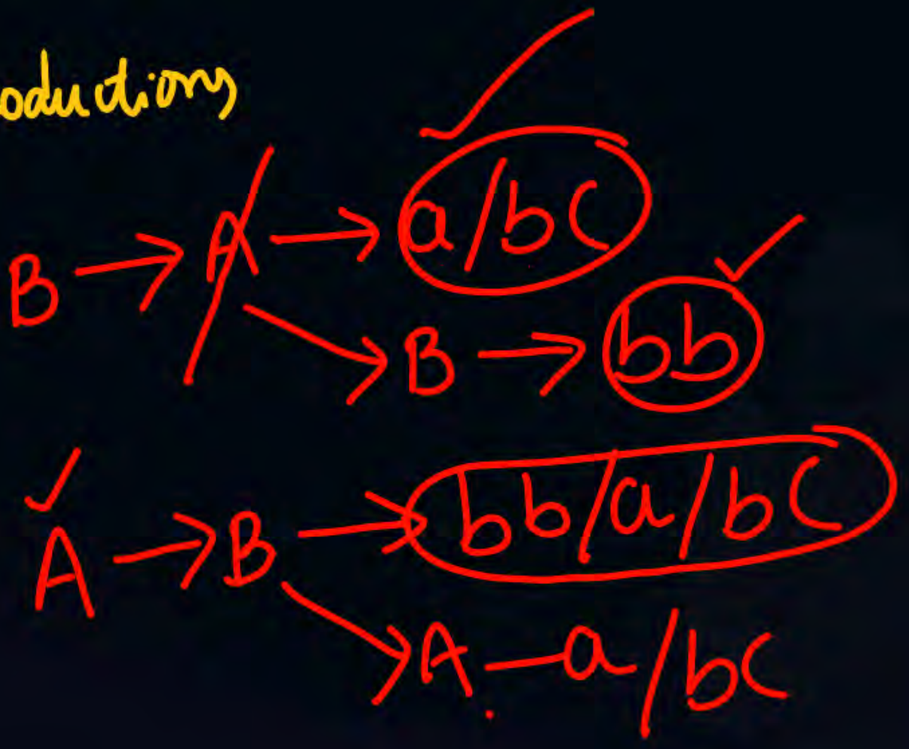
$S \to Aa / B / bb / a / bc$

$B \to A / bb / a / bc$

$A \to a / bc / B / bb$

Find all chains by unit productions





$S \to \check{A}\check{B}$

$A \to a$

$B \to C/b/a$

$C \to D/a$

$D \to a$

$E \to a$

$B \to C \to D \to E \to a$

$C \to D \to E \to a$

$D \to E \to a$

Simplification of CFG

1) ε $\in$

2) ε U

3) ε useless

11:33

11:38

→ Let R_1 and R_2 be regular

(a) $R_1 \cap R_2$ is not regular ✗

(b) $R_1 \cup R_2$ " ✗

(c) $\Sigma^* - R_1$ is regular ✓

(d) R_1^* is not regular ✓

Regular → $\cup \; \cap \; c \; *$

$\Sigma^* - R_1 = \overline{R_1}$

let $r = 1(1+0)^*$ $s = 11^*0$ $t = 1^*0$

a) $L(s) \subseteq L(r)$ and $L(s) \subseteq L(t)$

b)

c) $r = \{0, 10, 11, 100, 111, 110, 110\}$

d) $s = \{10, 110, 1110 \cdots\}$

$t = \{0, 10, 110, 1110 \cdots\}$

$s \subseteq t$ $r \supseteq s$

T/F

A FSM (Finite state machine) can be designed to add two integers of any arbitrary length

False

+, −

finite → long finite → Reg

The regular expression for the language recognized by the following machine



0^*1^*

0^* 1^* 0^*1^*

✓

Which of the following definitions generate the same language as L

where $L = \{x^n y^n / n \geq 1\}$

(i) $E \rightarrow xEy / xy$

(ii) $xy + x^+ x\, yy^+$

(iii) $x^+ y^+$



Which of the following are equivalent

(i) $(00)^*(\epsilon + 0)$ (ii) $(00)^*$ (iii) 0^* (iv) $0(00)^*$

(i) $(00)^*(\epsilon + 0)$ → Even and odd all 0's

(ii) $(00)^*$ → Even

(iii) 0^* → all 0's

(iv) $0(00)^*$ → Odd

$(00)^* + (00)^*0$ = all 0's $= 0^*$

$(00)^*$ → Even

$(00)^*0$ → odd

$(00)^*(\epsilon + 0)$

$(00)^*\epsilon + (00)^*0$

$(00)^*$ → Even ∪ $(00)^*0$ → odd

all zeros

Which of the following is true

a) $L = \{x / x$ has equal no of a's and b's$\}$ is regular

b) $L = \{a^n b^n / n \geq 1\}$ is regular

c) $L = \{x /$ x has more a's than b's$\}$ is regular

d) $L = \{a^m b^n / m, n \geq 1\}$ is regular

$a^+ b^+ \rightarrow RE$

$\downarrow$

Reg

→ which of the following RE's over (0,1) denotes set of all strings not containg (100) as substring

(a) $0^*(1+0)^*$ (b) $0^*(101)0^*$ (c) $0^*1^*01^*$

(d) $0^*(10+1)^*$

$0^*1^*01^*$

'ϵ' → is in language

ϵ

$L = \{\epsilon\}$

$A = (01+1)^*$

$B = ((01)^* 1^*)^*$

$(a+b)^* = (a^* b^*)^*$

$a \to 01$, $b \to 1$, $+ \to \cup$

a) $A \subset B$

b) $B \subset A$

c) A and B are incomparable

d) $A = B$

Which of the following can be recognised by DFA

a) The numbers 1, 2, 4, 8 ... ... 2^n ---- written in binary

X b) The " " " " " " unary

X c) The set of binary strings with equal no of zeros and ones

d) The set { 101, 11011, 1110111, 111101111, ... }



1, 10, 100, 1000, ...







1011



The string 1101 does not belong to the set represented by

a) $11(0+1)^*$ b) $1(0+1)^*101$

c) $(10)^*(01)^*(00+11)^*$ d) $(00+(11)^*0)^*$

11 01

THANK - YOU